

**SCHOOL OF
COMPUTING**

Amrita Vishwa Vidyapeetham, Chennai Campus
Department of Computer Science and Engineering
Exercise-3: Decision Tree Classifier

Name : M Dheeraj vamsi krishna

Roll no : CH.SC.U4CSE24127

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Your book's exercise describes a classification tree with age and income to determine whether someone buys a house. It states that age becomes the first node when the training data shows that 70% of people over age 30 bought a house.

```
import pandas as pd

from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import accuracy_score
from sklearn.metrics import confusion_matrix

import seaborn as sns
import matplotlib.pyplot as plt

data = {
    'Age': [22, 25, 28, 31, 35, 40, 45, 50, 55, 60,
            23, 27, 33, 37, 42, 48, 52, 58, 30, 65],
    'Income': [25, 30, 35, 40, 45, 50, 55, 60, 65, 70,
              28, 32, 42, 48, 52, 58, 62, 68, 38, 75],
    'BuysHouse': [0, 0, 0, 1, 1, 1, 1, 1, 1, 1,
                  0, 0, 1, 1, 1, 1, 1, 1, 1, 1]
}

df = pd.DataFrame(data)

print(df)
```

	Age	Income	BuysHouse
0	22	25	0
1	25	30	0
2	28	35	0
3	31	40	1
4	35	45	1
5	40	50	1
6	45	55	1
7	50	60	1
8	55	65	1
9	60	70	1
10	23	28	0
11	27	32	0
12	33	42	1
13	37	48	1
14	42	52	1
15	48	58	1
16	52	62	1
17	58	68	1
18	30	38	1
19	65	75	1

```

X = df[['Age', 'Income']]
y = df['BuysHouse']

X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.2,
    random_state=42
)

model = DecisionTreeClassifier(
    criterion='entropy',
    random_state=42
)

model.fit(X_train, y_train)

DecisionTreeClassifier(criterion='entropy', random_state=42)

y_pred = model.predict(X_test)

print("Actual:")
print(y_test.values)

print("\nPredicted:")
print(y_pred)

Actual:
[0 1 1 0]

Predicted:
[0 1 1 0]

accuracy = accuracy_score(y_test, y_pred)

print("Accuracy:", accuracy)
print("Accuracy Percentage:", accuracy * 100, "%")

Accuracy: 1.0
Accuracy Percentage: 100.0 %

cm = confusion_matrix(y_test, y_pred)

print("Confusion Matrix:")
print(cm)

Confusion Matrix:
[[2 0]
 [0 2]]

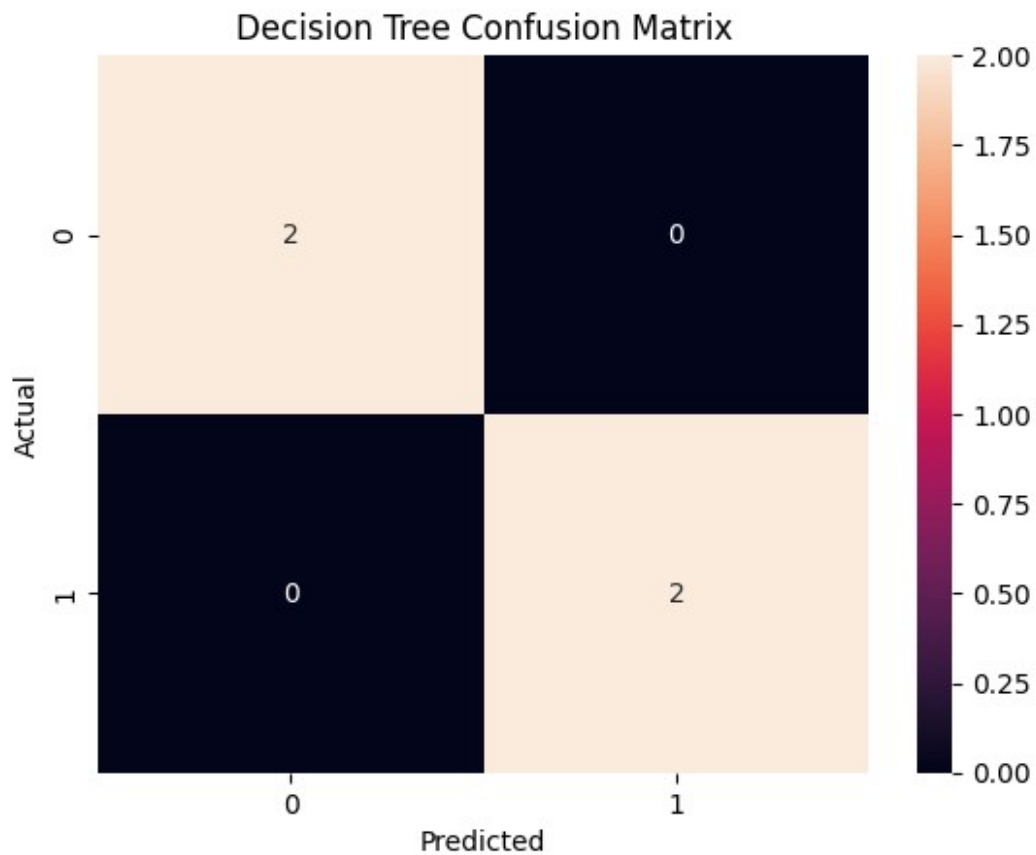
sns.heatmap(
    cm,
    annot=True,

```

```

    fmt='d'
)
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.title("Decision Tree Confusion Matrix")
plt.show()

```



Conclusion: A Decision Tree classifier was developed to predict whether a person buys a house using age and income. The model was trained successfully and evaluated using accuracy and a confusion matrix.